

Haeryun Jung

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EDUCATION

Ph.D. Candidate (ABD) in Sports Biomechanics

2020.03 - 2023.08

Kookmin University, Seoul, South Korea

- Sports Biomechanics & Sports Engineering Lab

M.S. in Sports Engineering Convergence

2015.03 - 2017.02

Kookmin University, Seoul, South Korea

- Sports Biomechanics & Sports Engineering Lab

B.S. in Adapted Physical Education

2011.03 - 2015.02

Youngin University, Yongin, South Korea

WORK EXPERIENCE

XR Developer

MTVS / Spatial X

Spatial X – XR Content for Disability Awareness through Spatial Perception Technology

- Developed immersive XR content for disability awareness using **Unity and C#**.
- Designed interactive experiences incorporating **spatial audio, spatial perception, and user interaction**.
- Implemented XR-based training and assessment features to provide users with an experiential understanding of sensory and spatial challenges.

Researcher

2023.08 - 2026.03

Korea Institute of Sport Science (KISS), Seoul, South Korea

Sports Biomechanics & Performance Analysis Team, Sports Science Research Center

- Conducted **performance analysis and performance prediction** for elite Paralympic athletes.
- Performed biomechanical analyses across multiple para-sports to support athlete performance and coaching.
- Developed and applied **sports science technologies and data-driven solutions** for Paralympic training and competition.
- Supported field-based performance monitoring, biomechanical assessment, and technology implementation for national-level para-athletes.

Research Assistant

2020.03 - 2023.08

Kookmin University, Seoul, South Korea

Sports Biomechanics & Sports Engineering Laboratory

- Collected, processed, and analyzed **biomechanical and human movement data**.
- Conducted **motion capture, functional assessment, and sports performance analysis** projects.
- Participated in **sports engineering and Sports-IT convergence research**, integrating biomechanics with sensing and digital technologies.
- Supported experimental design, data acquisition, signal processing, and quantitative analysis for human movement research.

PUBLICATIONS

Journal Papers

1. I.P. Hong, J. Yoo, B.K. Choi, T. Kim, **H.R. Jung**, G. Kim, S. Kim, K.H. Han, D. Kim

"Gender Differences in Postural Stability of Individuals with Intellectual Disabilities: A Static Balance Assessment (A Pilot Study)"

Korean Journal of Applied Biomechanics, 2025 | KCI | Vol. 35(1), pp.40-48 | ISSN: 1226-2226

[\[Paper\]](#)

2. E.H. Kwak, S.Y. Lee, **H.R. Jung**

"Effects of a Core Exercise Program on the Biomechanical Variables of Gait in Children with Intellectual Disability"

The Korea Journal of Sport, 2025 | KCI | Vol. 23(3), pp.311-320 | pISSN: 1738-3250

[\[Paper\]](#)

3. Y.H. Kim, S.H. Cho, **H.R. Jung**, K.K. Lee
"Method of Analyzing Important Variables using Machine Learning-based Golf Putting Direction Prediction Model"
 Korean Journal of Applied Biomechanics, 2022 | KCI | Vol. 32(1), pp.1-8 | ISSN: 1226-2226
[\[Paper\]](#)
4. W.J. Lee, **H.R. Jung**, S.J. Kong, S.G. Yoo, K.K. Lee
"The strategy of force production to control of amount of spin in preparation motion for vertical jump aerial twist movement"
 Korean Journal of Sport Science, 2022 | KCI | Vol. 31(1), pp.871-879
[\[Paper\]](#)
5. G. Ozkaya, **H.R. Jung**, I.S. Jeong, M.R. Choi, M.Y. Shin, X. Lin, W.S. Heo, M.S. Kim, E.H. Kim, K.K. Lee
"Three dimensional motion capture data during repetitive overarm throwing practice"
 Scientific Data, 2018 | SCIE | IF: 5.929 | Vol. 5, p.180272 | ISSN: 2052-4463
[\[Paper\]](#)
6. G. Ozkaya, **H.R. Jung**, I.S. Jeong, M.R. Choi, M.Y. Shin, X. Lin, W.S. Heo, M.S. Kim, K.K. Lee
"Relationship between the Ball Velocity and Upper Extremity Kinematic Variables during an Overarm Throwing Task of Inexperienced Individuals"
 Korean Journal of Sports Biomechanics, 2017 | KCI | Vol. 27(1), pp.19-23 | ISSN: 1226-2226
[\[Paper\]](#)
7. M.S. Kim, K.K. Lee, **H.R. Jung**, Y.S. Jee, K.C. Lim
"Effect of the beat on Nanta player's motion"
 The Official Journal of the Korean Academy of Kinesiology, 2016 | KCI | Vol. 18(4), pp.45-52 | ISSN: 2586-5552

Conference Papers & Submissions

1.
"Peak Performance Age in Parasports: Kaplan-Meier Survival Curves"
 VISTA 2025, International Paralympic Committee (IPC) | Conference submission | Submitted | Conference held online May 18-28, 2026
[\[Paper\]](#)
2. **H.R. Jung**, K.K. Lee
"Glove type and Spatio-Temporal variables of Wheelchair Racer according Paralympic Sports Classification in 100m Sprint"
 6th International Rehabmove State-of-the-Art Congress, 2018 | Proceeding | ISBN: 978-90-75823-93-6
[\[Paper\]](#)
3. **H.R. Jung**, N.R. Kang
"Paralympic Sports Classification and Spatio-Temporal parameter of Wheelchair Racer in Male's 100m Sprint"
 6th International Rehabmove State-of-the-Art Congress, 2018 | Proceeding | ISBN: 978-90-75823-93-6
[\[Paper\]](#)
4. M.Y. Shin, K.K. Lee, **H.R. Jung**, M.R. Choi
"Posture Stability According to Arch Height Using Depth Camera"
 29th 88 Seoul Olympic Anniversary International Sports Science Congress, 2017 | Proceeding | pp.242
5. **H.R. Jung**, S.J. Kim, K.K. Lee
"The laterality for right handed and left handed in rhythmic bimanual coordination"
 7th Asia Conference on Kinesiology, 2016 | Proceeding | pp.108-109
6. **H.R. Jung**, K.K. Lee, S.J. Kim, M.R. Choi, J.H. Lee
"Qualitative comparison of Disc movement and Inter joint coordination of upper limb in Flying Disc Forehand throwing"
 34th International Conference on Biomechanics in Sports, 2016 | Proceeding | 2016
[\[Paper\]](#)

RESEARCH EXPERIENCE

Biomechanics Measurement & Analysis Service	2020.02 - 2020.11
SK Wyverns Co., Ltd. Kookmin University	
Smart Electronic Judging System for Practical Taekwondo Competition	2020.06 - 2021.01
Ministry of Culture, Sports and Tourism Kookmin University	

Biomechanical Running/Gait Posture Analysis for Hearable Coaching & Body Age Evaluation Device Ministry of Culture, Sports and Tourism Kookmin University	2020.07 - 2020.12
2020 Basic Sports (Athletics, Swimming, Gymnastics) Video Analysis Support Ministry of Culture, Sports and Tourism Kookmin University	2020.10 - 2021.02
Smart Insole Parameter Extraction & Motion Analysis Algorithm Development Solted Co., Ltd. Kookmin University	2021.01 - 2021.10
Smart Electronic Judging System for Practical Taekwondo Competition (2nd Year) Ministry of Culture, Sports and Tourism Kookmin University	2021.02 - 2022.01
Modular Smart Fashion Platform Research Center (Phase 3, 2/2) Ministry of Science and ICT Kookmin University	2021.03 - 2022.02
Interactive Healthcare Smart Wear & Service Platform with Integrated Motion Analysis (1st Year) Ministry of Trade, Industry and Energy Kookmin University	2021.04 - 2021.12
Rehabilitation & Leisure Sports Linked Exercise Data Analysis and Verification Support (Phase 1, 1/2) Ministry of Science and ICT Kookmin University	2021.04 - 2021.08
Sports Sock Functionality Test Protocol Development Ministry of Education Kookmin University	2021.06 - 2021.12
Smart Clothing Technology-based Motion Analysis & Exercise Estimation for Fitness Program Contents (1/4) Ministry of Education Kookmin University	2021.09 - 2022.03
Rehabilitation & Leisure Sports Linked Exercise Data Analysis and Verification Support (Phase 1, 2/2) Ministry of Science and ICT Kookmin University	2022.02 - 2022.08
Interactive Healthcare Smart Wear & Service Platform with Integrated Motion Analysis (2nd Year) Ministry of Trade, Industry and Energy Kookmin University	2022.02 - 2022.08
Effect of Mediaid Assist Gear on Upper Limb Motion and Lumbar EMG During Heavy Box Lifting Industry Partner Kookmin University	2022.04 - 2022.05

SKILLS

Hardware

- 3D Motion Capture System, EMG (Electromyography), Pressure Measurement System, Force Plate, IMU/GPS System, Sports Video Analysis System

Software

- C#, Python, MATLAB (Data Processing, Statistical Analysis, Visualization), OpenCV, Unity, Biomechanics Analysis Software